

Systematic Map Codebook — D5.6

ILRI / Bristlepine Resilience Consultants — 2025–2026

Scope: Systematic map data extraction only (D4/D5). Not the systematic review (D8/D9).

The 16 coding fields

Use this as your reference while coding. Valid values must be used exactly as written (except free-text fields, marked △).

1. `publication_year`

Year the study was published. Use the most recent version date if both preprint and journal exist. **Valid values:** integer, e.g. `2021`

2. `publication_type`

Value	Meaning
<code>journal_article</code>	Peer-reviewed journal (including open-access)
<code>report</code>	Project/programme report, evaluation report, policy brief
<code>working_paper</code>	Pre-publication working paper or discussion paper
<code>thesis</code>	PhD or master's dissertation
<code>other</code>	Book chapter, conference paper, dataset paper

3. `country_region`

Country or countries where the study was conducted. Use region label only if no countries are named. **Valid values:** free text, semicolons between multiple entries

Examples: `Tanzania` | `Ghana; Mali; Burkina Faso` | `East Africa` | `Global`

4. `geographic_scale`

Value	Meaning
<code>local</code>	Village, community, single site
<code>sub-national</code>	District, province, region within a country
<code>national</code>	One country
<code>multi-country</code>	Two or more specific named countries
<code>regional</code>	Broad region without country-level focus

5. `producer_type`

Select all that apply, semicolons between multiple values.

Value	Meaning
crop	Crop farming systems
livestock	Livestock farming systems
fisheries_aquaculture	Fisheries or aquaculture systems
agroforestry	Agroforestry systems
mixed	Explicitly mixed crop-livestock
undefined	Generic “smallholder farmers” with no further system specified

6. `marginalized_subpopulations`

Code only what is explicitly stated — do not infer. Select all that apply.

Value	Meaning
women	Women explicitly targeted or gender-disaggregated analysis
youth	Young farmers or age-disaggregated outcomes
people_with_disabilities	Persons with disabilities explicitly named
landless	Landless households or tenure-insecure groups
indigenous_peoples	Indigenous communities named
ethnic_minorities	Ethnic minority groups named
migrant_seasonal_workers	Migrant or seasonal agricultural workers
other	Any other marginalised group explicitly named (e.g. caste-based groups, religious minorities) — add a brief note in the <code>notes</code> field
none_reported	No marginalised groups mentioned

7. `adaptation_focus` *free text — emergent*

The specific climate adaptation action, intervention, or practice the study tracks. 5–15 words.

Examples: `crop management practices` | `crop varieties and genetics` | `water management and irrigation` | `soil and land management` | `agroforestry` | `livestock management` | `livelihood diversification (non-farm)` | `financial strategies` | `climate information and advisory services` | `migration and mobility` | `social and community networking` | `post-harvest and consumption adjustments`

8. `process_outcome_domains`

Select all that apply, semicolons between multiple values.

Code	Meaning
knowledge_awareness_learning	Awareness of climate change, knowledge of options, learning outcomes
decision_making_planning	Farmer decision-making, use of forecasts in planning
uptake_adoption	Rates of adoption of a practice or technology
behavioral_change	Changes in farming or land management behaviour
participation_coproduction	Participation in programmes, co-design, farmer group engagement
institutional_governance	Policy uptake, institutional change, governance
access_information_services	Access to extension, credit, markets, climate information
yields_productivity	Crop yield, livestock productivity, fish catch
income_assets	Household income, asset accumulation
livelihoods	Food security, livelihood diversification
wellbeing	Health, nutrition, subjective wellbeing
risk_reduction	Reduced exposure or sensitivity to climate shocks
resilience_adaptive_capacity	Adaptive capacity indices, resilience scores, vulnerability indices

9. indicators_measured free text

The specific indicators or metrics used. Extract actual indicators from methods/results. Include units.

Examples: yield (t/ha), income (USD/season), food security score (HFIAS) | adoption rate (%), area under improved variety (ha)

10. methodological_approach

Select all that apply, semicolons between multiple values. For explicitly mixed-methods studies: select both qualitative and quantitative.

Value	Meaning
qualitative	Qualitative research design (interviews, focus groups, ethnography)
quantitative	Quantitative research design (surveys, statistical analysis)
participatory	Only when participation IS the primary design (PRA, photovoice) — not when farmers are merely surveyed
modeling_with_empirical_validation	Crop/climate/agent-based models validated with field data
experimental	RCTs, natural experiments, quasi-experimental approaches — purely lab/field experiments without social application excluded

11. purpose_of_assessment

Value	Meaning
research	Academic study with no programme affiliation
project_learning	M&E embedded in an ongoing project
program_evaluation	Formal evaluation of a completed programme
donor_reporting	Report produced for a donor/funder
national_reporting	Contributing to NAPs, NDCs, national monitoring frameworks

12. data_sources

Select all that apply.

Value	Meaning
surveys	Primary household or farm surveys
administrative_data	Government or programme records
remote_sensing	Satellite or aerial imagery
participatory_methods	PRA, focus groups, photovoice as a data source
secondary_data	Existing datasets, literature, or other secondary sources
other	Any data source not listed above

13. temporal_coverage

Value	Meaning
cross_sectional	Single point in time or single season
seasonal	Intra-annual, across one or a few seasons
longitudinal	Multi-year, panel data, repeated measures with the same individuals
repeated_cross_sectional	Multi-year, repeated measures with different individuals/units each time

14. cost_data_reported

Value	Meaning
yes	Paper explicitly reports cost figures or cost-effectiveness ratios
no	No cost data reported — qualitative mention of “limited resources” counts as no

15. strengths_and_limitations free text

Author-reported strengths and limitations combined (protocol Table 3). Label each component.

Format: Strength: <text>. Limitation: <text>. Look in Discussion, Conclusion, and Limitations sections. Extract verbatim where possible.

16. lessons_learned free text

Key lessons or recommendations the authors report. 1–3 sentences. Focus on what they say should be done differently — not findings. Enter not_reported if absent.

Reconciliation (Step 2)

Initial round (FT-R1a, 5 papers) — full reconciliation:

1. Open both sheets side by side
2. For each paper and each field:
 1. Agree → enter that value in reconciled_ft_r1a.csv
 2. Disagree → discuss until you reach agreement, then enter it
3. Record the reason for any substantive disagreement in reconciliation_notes
4. Cannot agree → escalate to lead researcher

Do not average numeric fields — always discuss to a single agreed value.

Subsequent rounds — random subset convergence:

Each coder independently codes their assigned papers. After each round, a random subset is compared across coders to check agreement. The goal is convergence — when agreement is stable across two consecutive subsets, coding is considered consistent. Subset size agreed with the lead researcher before each round.

Inter-rater reliability (IRR)

After reconciliation, run:

```
python scripts/step11_irr_analysis.py \  
--h1 documentation/coding/systematic-map/rounds/FT-R1a/coding_H1.csv \  
--h2 documentation/coding/systematic-map/rounds/FT-R1a/coding_H2.csv \  
--gold documentation/coding/systematic-map/rounds/FT-R1a/reconciled.csv \  
--llm documentation/coding/systematic-map/rounds/FT-R1a/coding_LLM.csv \  
--out documentation/coding/systematic-map/rounds/FT-R1a/irr_R1a.json
```

Kappa interpretation

κ	Interpretation	Action
< 0.20	Slight	Major revision — field may need redesign
0.21–0.40	Fair	Significant revision — add examples, tighten definitions
0.41–0.60	Moderate	Revision needed — review disagreement patterns
0.61–0.80	Substantial	Approaching threshold — fine-tuning only
> 0.80	Almost perfect	Ready to lock

Threshold to lock a closed field: $\kappa \geq 0.60$ AND sensitivity ≥ 0.80 in the same batch.

Emergent field (`adaptation_focus`)

Tracked by category saturation, not kappa. Saturated when no new codes appear across two consecutive batches.

Stopping rules

- **Closed fields** — locked once $\kappa \geq 0.60$ AND sensitivity ≥ 0.80 in the same batch. Locked fields are not retested.
- **Emergent field** — `adaptation_focus` saturated when no new codes appear across two consecutive batches.
- **Full corpus run** — proceed to step15 once all closed fields are locked AND `adaptation_focus` is saturated.

How rounds work

Round naming

Label	Who codes	Papers
FT-R1a	H1 + H2 + LLM independently	First 5 papers
FT-R1b	LLM only (re-codes same R1a papers after criteria update)	Same 5 papers as R1a
FT-R2a	H1 + H2 + LLM independently	Next 5 papers (6–10)
FT-R2b	LLM only (re-codes R2a papers after criteria update)	Same 5 as R2a

a rounds = full human + LLM pass on new papers. **b rounds** = LLM verification only, same papers, after criteria update. Human coders only ever code in **a** rounds.

Each round — six steps

```
Step 1 – Code independently  
H1, H2, and LLM all code the same 5 papers. No discussion.  
↓
```

Step 2 – Reconcile (H1 + H2 only)

Compare field by field. Agree on gold standard → reconciled.csv

↓

Step 3 – Compute IRR

Run step11. Kappa + sensitivity per field and overall.

↓

Step 4 – Review disagreements

Which fields underperformed? Was guidance unclear? Categories missing?

↓

Step 5 – Update codebook and LLM criteria

Update CODEBOOK.md with clarified definitions and new examples.

Copy llm-criteria/criteria_vN.yml → criteria_v(N+1).yml with field-level improvements. Log the change in llm-criteria/CHANGELOG.md.

↓

Step 6 – Check stopping rules

All closed fields locked AND emergent fields saturated → run full corpus.

Otherwise → LLM re-codes same batch (FT-R1b) to verify improvement, then proceed to next round (FT-R2a) with updated criteria.

Folder structure

```
documentation/coding/systematic-map/
├── CODEBOOK.md           ← this document
├── llm-criteria/
│   ├── criteria_sysmap_v1.yml      ← machine-readable LLM prompt (loaded by step15)
│   ├── criteria_sysmap_v1.md      ← human-readable companion
│   └── CHANGELOG.md
├── rounds/
│   ├── FT-R1a/
│   │   ├── FT-R1a pdfs/           ← full-text PDFs/HTMLs (on Google Drive, not in git)
│   │   ├── coding_ft_r1a_XX.csv   ← blank template (coders save-as coding_ft_r1a_CS.csv etc.)
│   │   ├── reconciled_ft_r1a.csv  ← gold standard after reconciliation
│   │   └── irr_ft_r1a.json        ← kappa, sensitivity, field-level stats (generated)
│   │   [on Drive only: coding_ft_r1a_CS.csv, coding_ft_r1a_JD.csv, coding_ft_r1a_LLM.csv]
│   ├── FT-R1b/                  ← LLM re-code only (same papers as R1a)
│   ├── FT-R2a/                  ← next batch
│   └── ...
documentation/coding/abstract-screening/
├── criteria/                   ← eligibility criteria used in abstract screening (locked)
```

Round log

Round	Papers	Date	κ (subset)	Status
FT-R1a	5	—	—	🕒 In progress
FT-R2a	5	—	—	— Not started

Update after each round's IRR run.

Quick reference — valid values

Field	Values
publication_type	journal_article, report, working_paper, thesis, other
geographic_scale	local, sub-national, national, multi-country, regional
producer_type	crop, livestock, fisheries_aquaculture, agroforestry, mixed, undefined
marginalized_subpopulations	women, youth, people_with_disabilities, landless, indigenous_peoples, ethnic_minorities, migrant_seasonal_workers, other, none_reported
adaptation_focus	free text
process_outcome_domains	knowledge_awareness_learning, decision_making_planning, uptake_adoption, behavioral_change, participation_coproduction, institutional_governance, access_information_services, yields_productivity, income_assets, livelihoods, wellbeing, risk_reduction, resilience_adaptive_capacity
indicators_measured	free text
methodological_approach	qualitative, quantitative, participatory, modeling_with_empirical_validation, experimental
purpose_of_assessment	project_learning, program_evaluation, donor_reporting, national_reporting, research
data_sources	surveys, administrative_data, remote_sensing, participatory_methods, secondary_data, other
temporal_coverage	cross_sectional, seasonal, longitudinal, repeated_cross_sectional
cost_data_reported	yes, no
strengths_and_limitations	free text (Strength: ... / Limitation: ...)
lessons_learned	free text

Note for lead researcher — LLM parallel workflow

In addition to the human coding rounds described above, a large-language model (LLM) independently codes the full corpus using the same schema (`llm-criteria/criteria_sysmap_v1.yml`). This runs in the background and is invisible to human coders.

The LLM output serves as a secondary quality signal: after each human round, agreement rates between human codes and LLM codes are computed alongside inter-rater statistics. This does **not** replace human reconciliation — it supplements it. Human agreement on the random subset remains the primary stopping criterion.

LLM criteria are versioned (`criteria_sysmap_v1.yml` , `criteria_v2.yml` , ...) and updated after each round's review. Changes are logged in `llm-criteria/CHANGELOG.md` .